



Dyeing chemicals

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▼ 1. Introduction

Wet processing stage is one of the significant processes involving chemical usage in textiles. Speciality chemicals and dyes are used across pretreatment, dyeing and post treatment stages and this continues into the finishing stages.

Goal and scope definition

The objective of this work is to develop Life Cycle Inventory (LCI) datasets for textile specialty chemicals and dyes that can serve as robust alternatives to the Environmental Footprint (EF) 3.1 datasets, which are currently widely used but limited in granularity and transparency for textile wet processing applications.

The **functional unit** is defined as:

1 kg of textile chemical or dye, cradle-to-gate, ready to use in textile dyeing or wet processing.

The **system boundary** covers:

- Raw material extraction and upstream chemical precursors
- Chemical synthesis and processing
- Formulation and blending
- Energy, water, and auxiliary inputs

- On-site emissions and waste treatment up to factory gate

▼ 2. Data sources and methodology

The methodology is designed recognising that textile chemicals and dyes are among the most **black-boxed processes** in industrial LCA due to confidentiality, proprietary formulations, and supplier-specific synthesis routes.

2.1 Data sources

Data collection was carried out using the following sources:

1. Peer-reviewed journals and technical literature

Used to identify:

- Typical synthesis routes for dye classes and textile auxiliaries
- Energy and material intensity ranges
- Reaction yields and by-products

2. Ecoinvent database

Used as a background database for:

- Upstream raw materials (e.g. basic organic chemicals, acids, alkalis)
- Energy supply, utilities, and waste treatment

It is to be noted that ecoinvent represents most textile dyes and auxiliaries using generic chemical datasets, often derived from an average of approximately 20 commonly used organic/inorganic chemicals. This approach masks chemical-specific differences and introduces uncertainty when applied to textile wet processing.

3. Publicly available supplier information

Special emphasis was placed on:

- MSDS sheets
- Technical datasheets

Websites of major global textile chemical and dye suppliers (e.g. Archroma and similar top-tier suppliers) were systematically reviewed. MSDS documents were

analysed to identify chemical composition ranges, hazardous constituents, solvents, and stabilisers disclosed under regulatory requirements.

2.2 Two-part data collection framework

LCI development for textile chemicals and dyes was structured into two complementary components:

Chemical and dye production (foreground supply)

This component focuses on the production of 1 kg of chemical or dye, including:

- Identification of dominant raw materials and intermediates
- Mapping of typical synthesis pathways (where available)
- Estimation of energy and utility requirements
- Allocation of emissions and wastes at the production stage

Where exact synthesis routes are unavailable due to confidentiality, chemically similar proxy substances were selected based on molecular structure, functional groups, and reaction complexity.

This step ensures comparability with EF-style datasets while improving chemical specificity wherever possible.

Use-phase relevance: dosage, concentration, and pick-up

To ensure usability in textile LCAs, production LCIs were complemented with process-relevant parameters, including:

- Typical dosage ranges (e.g. g/kg fabric or % on weight of fabric (OWF), meaning the chemical amount is expressed as a percentage of the fabric's dry weight)
- Concentration levels in dye baths or finishing liquors
- Pick-up ratios and fixation efficiencies
- Fate of chemicals (fixed to fabric vs discharged to wastewater)

This allows the developed LCIs to be scaled from “per kg chemical” to “per kg fabric processed”, bridging the gap between chemical production inventories and textile process inventories.

Where applicable, differences between fibre types (e.g. cotton, polyester, wool) and dyeing technologies (batch, continuous, semi-continuous) were considered.

2.3 Addressing data gaps and “black-box” challenges

Textile specialty chemicals and dyes present several structural challenges for LCI development:

- **Lack of transparent production data** due to proprietary synthesis routes
- **Multiplicity of chemicals for the same function**, where several chemically distinct substances can perform identical roles (e.g. levelling agents, dispersing agents)
- **Aggregation in existing databases**, where dyes, auxiliaries, and finishing agents are represented by generic organic chemical averages

To address these issues, the following measures were applied:

- Only chemicals and dyes for which minimum composition or structural information is available in the public domain were included in the initial dataset development.
- Chemicals were classified by function, not by brand, allowing aggregation where appropriate while retaining functional relevance.
- Where multiple chemicals serve the same function, the methodology allows for future differentiation rather than forcing a single representative value.

▼ 3. Life Cycle Inventory Modelling

In this section, all assumptions used to model LCI of chemicals/dyes are explored.

3.1 Specialty chemicals

Specialty chemical inventories were modelled per kg of chemical produced, cradle-to-gate, using [ecoinvent 3.12](#) background data. For most textile specialty chemicals, a suitable existing ecoinvent 3.12 market activity was identified and used directly as the inventory — the mapping from each chemical function to the corresponding ecoinvent dataset is given in Section 4.1. For the remaining entries where no single ecoinvent dataset represents the formulation (e.g. salt mixes, water-based dilutions, multi-component blends), a custom composition was built

from ecoinvent components. Climate change impacts (kg CO₂eq/kg chemical) were considered for 24 chemical function categories (wetting agents, scouring agents, electrolytes, dispersing agents, levelling agents, fixing agents, pH control agents, softeners, etc.).

Chemical dosing rules differ by process type:

- **Exhaust dyeing:** chemical recipes are expressed in g/L. A default liquor ratio is assumed per substrate/technology/dye type combination, determining absolute chemical quantities per kg of textile.
- **Continuous and CPB dyeing:** chemical recipes are expressed in % on weight of fabric (OWF). The pick-up rate determines the actual chemical load.

A fixed repartition of the chemical inventory is applied within each process category, while the absolute quantity scales with the liquor ratio or pick-up rate.

3.2 Dyestuff

Since detailed dye composition and production data are not publicly available, dye LCIs were developed using a bottom-up, industry-informed approach. For each dye class, a representative production route was identified from environmental clearance applications and literature, and the inventory was constructed using ecoinvent proxy datasets for intermediates with comparable chemistry.

Detailed LCI along with assumptions can be found in the annex.

▼ 4. Life Cycle Impact Assessment

In this section, LCIA results of speciality chemicals and dyes are discussed.

4.1 Speciality chemicals

The results indicate a wide variation in climate impacts across speciality chemicals used in wet processing, spanning roughly an order of magnitude between the lowest-impact basic processing aids and the highest-impact complex organic formulations. Specific per-chemical values are not reproduced here, as most of the underlying impacts are taken directly from ecoinvent 3.12 datasets; readers should consult ecoinvent via the dataset links in the mapping table for values.

Function	Chemical	Composition / Dataset	Climate change (kg CO ₂ e/kg)
Anti slippage agent	Non ionic surfactant	<u>market for non-ionic surfactant</u> <u>non-ionic surfactant</u> <u>GLO</u> kg	-
Anti-migrating agent	Polyacrylates based soaping agent	<u>market for polyacrylamide</u> <u>polyacrylamide</u> <u>GLO</u> kg	-
Auxiliary	Napthalene sulfonic acid	<u>market for naphthalene sulfonic acid</u> <u>naphthalene sulfonic acid</u> <u>GLO</u> kg	-
Bleaching agent	Hydrogen peroxide	<u>market for hydrogen peroxide, without water, in 50% solution state</u> <u>hydrogen peroxide, without water, in 50% solution state</u> <u>RoW</u> kg	-
Chelating / Sequestering agent	EDTA	<u>market for EDTA, ethylenediaminetetraacetic acid</u> <u>EDTA, ethylenediaminetetraacetic acid</u> <u>GLO</u> kg	-
De-foaming agent	Polysiloxane	<u>market for polydimethylsiloxane</u> <u>polydimethylsiloxane</u> <u>RoW</u> kg	-
Dispersing agent	Alcohol ether sulphate	<u>market for fatty alcohol sulfate</u> <u>fatty alcohol sulfate</u> <u>GLO</u> kg	-
Dispersing agent	Acrylic dispersion	<u>market for acrylic dispersion, with water, in 58% solution state</u> <u>acrylic dispersion, with water, in 58% solution state</u> <u>RoW</u> kg	-
Electrolyte	Salt mix	Sodium chloride + sodium sulfate + potassium chloride	-
Emulsifier / detergent	Non ionic surfactant	<u>market for non-ionic surfactant</u> <u>non-ionic surfactant</u> <u>GLO</u> kg	-

Exhausting agent	Sodium chloride	<u>market for sodium chloride, powder sodium chloride, powder GLO kg</u>	-
Fixing agent	Benzisothiazolinone based	Production mix + methallyl chloride	-
Fixing agent	Acetic acid	<u>market for acetic acid acetic acid GLO kg</u>	-
Peroxide killer	Enzymes	<u>market for enzymes enzymes GLO kg</u>	-
Levelling agent 1	Non ionic surfactant	<u>market for non-ionic surfactant non-ionic surfactant GLO kg</u>	-
Levelling agent 2	Non ionic surfactant	<u>market for non-ionic surfactant non-ionic surfactant GLO kg</u>	-
Levelling agent 3	Enzymes	Water (95%) + enzymes (5%)	-
Levelling agent 4	Diphenyl ether	Diphenyl ether compound production	-
Levelling agent 5	Non ionic surfactant mix	EO + fatty acid derivative (50/50)	-
Lubricant	Polyacrylamide	<u>market for polyacrylamide polyacrylamide GLO kg</u>	-
Lubricant	Organophosphorus compound	<u>market for organophosphorus- compound, unspecified organophosphorus- compound, unspecified GLO kg</u>	-
Mild oxidising agent	Acrylic dispersion	<u>market for acrylic dispersion, with water, in 58% solution state acrylic dispersion, with water, in 58% solution state RoW kg</u>	-
Mild oxidising agent	Hydrogen peroxide	<u>market for hydrogen peroxide, without water, in 50% solution state hydrogen peroxide,</u>	-

		<u>without water, in 50% solution state RoW kg</u>	
Moisture management	Acrylic dispersion	<u>market for acrylic dispersion, with water, in 58% solution state acrylic dispersion, with water, in 58% solution state RoW kg</u>	-
Moisture management	Polyester resin	Polyester resin (15%) + water (85%)	-
Oligomer remover	Silicone emulsion	<u>market for silicone product silicone product RoW kg</u>	-
Peroxide killer	Catalase	<u>market for enzymes enzymes GLO kg</u>	-
pH control & neutralisation 1	Acetic acid + ammonium sulphate	Mix (25% + 25% + water)	-
pH control & neutralisation 2	Citric acid	<u>market for citric acid citric acid GLO kg</u>	-
pH control & neutralisation 3	Caustic soda	<u>market for sodium hydroxide, without water, in 50% solution state sodium hydroxide, without water, in 50% solution state RoW kg</u>	-
pH control & neutralisation 4	Acetic acid	<u>market for acetic acid acetic acid GLO kg</u>	-
pH control & neutralisation 5	Soda ash	<u>market for soda ash, light soda ash, light RoW kg</u>	-
pH control & neutralisation 6	Sodium carbonate	<u>market for soda ash, light soda ash, light RoW kg</u>	-
pH control & neutralisation 7	Sodium hydroxide	<u>market for sodium hydroxide, without water, in 50% solution state sodium hydroxide, without water, in 50% solution state RoW kg</u>	-
Polyethylene / silicone emulsion	Silicone emulsion	<u>market for silicone product silicone product RoW kg</u>	-

Polyethylene / silicone emulsion	Polysiloxane	<u>market for polydimethylsiloxane</u> <u>polydimethylsiloxane</u> <u>RoW</u> kg	-
Reduction agent 1	Sodium dithionite	<u>market for sodium dithionite, anhydrous</u> <u>sodium dithionite, anhydrous</u> <u>RoW</u> kg	-
Reduction agent 2	Sodium dithionite + formaldehyde	Mixed production	-
Reduction agent 3	Sodium hydrosulphide	<u>market for sodium hydrosulfide</u> <u>sodium hydrosulfide</u> <u>GLO</u> kg	-
Refiner	Ethoxylated alcohol	<u>market for ethoxylated alcohol (AE>20)</u> <u>ethoxylated alcohol (AE>20)</u> <u>GLO</u> kg	-
Salt	Sodium sulphate	<u>market for sodium sulfate, anhydrite</u> <u>sodium sulfate, anhydrite</u> <u>RoW</u> kg	-
Scouring & soaping 1	Ethoxylated alcohol mix	Water (70%) + AE7 (30%)	-
Scouring & soaping 2	Polyacrylate mix	Water (70%) + polyacrylamide (30%)	-
Scouring agent 1	Non ionic surfactant	<u>market for non-ionic surfactant</u> <u>non-ionic surfactant</u> <u>GLO</u> kg	-
Scouring agent 2	Polysiloxane	<u>market for polydimethylsiloxane</u> <u>polydimethylsiloxane</u> <u>RoW</u> kg	-
Scouring agent 3	Non ionic surfactant	<u>market for non-ionic surfactant</u> <u>non-ionic surfactant</u> <u>GLO</u> kg	-
Scouring agent 4	Polyacrylates	Polyacrylate in water	-
Scouring agent 5	Glycerol esters	GMS-based production	-

Softener 1	Polysiloxane	<u>market for polydimethylsiloxane polydimethylsiloxane RoW kg</u>	-
Softener 2	Soybean-based	Soybean meal + water	-
Softener 3	Ethoxylated alcohol	<u>market for ethoxylated alcohol (AE>20) ethoxylated alcohol (AE>20) GLO kg</u>	-
Stabilizer	Magnesium sulfate	<u>market for magnesium sulfate magnesium sulfate GLO kg</u>	-
Stiffener	PVOH	Polyvinyl acetate production	-
Wetting agent 1	Non ionic surfactant	EO + fatty acid derivative	-
Wetting agent 2	Non ionic surfactant	Mixed non-ionic system	-
Wetting agent 3	Ethoxylated alcohol	<u>market for ethoxylated alcohol (AE>20) ethoxylated alcohol (AE>20) GLO kg</u>	-

Lower-impact chemicals are predominantly basic processing aids such as salts, alkalis like sodium hydroxide and soda ash, enzymes, and peroxide stabilisers. In contrast, higher impacts are associated with more complex organic formulations, particularly polysiloxane-based products, diphenyl ether levelling agents, organophosphorus lubricants, and PVOH-based stiffeners. Surfactant-based chemicals, which are widely used across multiple stages (wetting, scouring, levelling, and soaping), show moderate to high impacts depending on their chemistry.

Functionally, the results highlight that chemicals used in finishing and performance enhancement—such as softeners, levelling agents, lubricants, and stiffeners—tend to have the highest per-kg climate impacts, while bulk process chemicals and auxiliaries required for bath chemistry control or basic reactions show significantly lower impacts. Intermediate categories such as dispersing agents, fixing agents, and reduction agents fall within a mid-range, reflecting a mix of relatively simple and more complex chemistries. Overall, the distribution suggests that while high-impact chemicals may be used in smaller quantities, they represent critical hotspots from a per-kg perspective within wet processing operations.

4.2 Dyes

Dye type	Production route
Azo acid dyes	Bromisation production
Anthraquinone vat dyes	Alkali fusion production
Azo disperse dyes	Diazotization–coupling production
Azo reactive dyes	Monochlorotriazine production

Among the four dye classes, azo acid dyes show the highest climate impact, driven primarily by the use of nitro-aromatic intermediates (e.g., 2-nitroaniline) and bromine chemistry, both of which are energy-intensive and fossil-based, contributing significantly to upstream emissions. Azo reactive dyes show the lowest impact: although still reliant on aromatic intermediates, their impact is more evenly distributed across multiple inputs, with comparatively lower reliance on highly intensive precursors.

Overall, raw material synthesis and process heat are the primary drivers of emissions for these dye systems, rather than downstream utilities or minor inputs.

▼ 5. Limitations and future work

- The carbon footprints of speciality chemicals are highly dependent on their underlying composition. However, these formulations are largely treated as black boxes, and proxies representing commonly used chemistries have been applied, which may not fully capture product-specific variations.
- Greater transparency on the exact composition of speciality chemicals used in wet processing would significantly improve accuracy, particularly for multi-component formulations where the dominant contributors are not always disclosed.
- For dyes, a bottom-up modelling approach has been adopted using indicative compositions from manufacturers, combined with proxy datasets for raw materials; this introduces uncertainty where actual supply chains or synthesis routes differ.
- Energy consumption in dye production has been estimated based on typical process routes (e.g., diazotization, coupling, alkali fusion), rather than plant-

specific data, which may not reflect operational efficiencies or regional differences.

- Future improvements can be achieved through the use of primary data from chemical and dye manufacturers, including detailed bill of materials, process-specific energy use, and site-level emissions, enabling more representative and robust footprint calculations.

▼ 5. References

5.1 Speciality chemicals

1. Roos, S., Jönsson, C., Posner, S. *et al.* An inventory framework for inclusion of textile chemicals in life cycle assessment. *Int J Life Cycle Assess* **24**, 838–847 (2019). <https://doi.org/10.1007/s11367-018-1537-6>
2. European Textile Chemical Corporation: <https://eurotexchem.com/>

5.2 Dyes

1. Acid dye: [Agrasen colours private limited](#)
2. Disperse dye: [Parishi chemicals private limited](#)
3. Vat dye: [Agrasen colours private limited](#)
4. Reactive dye: [Adityayam Industries private limited](#)

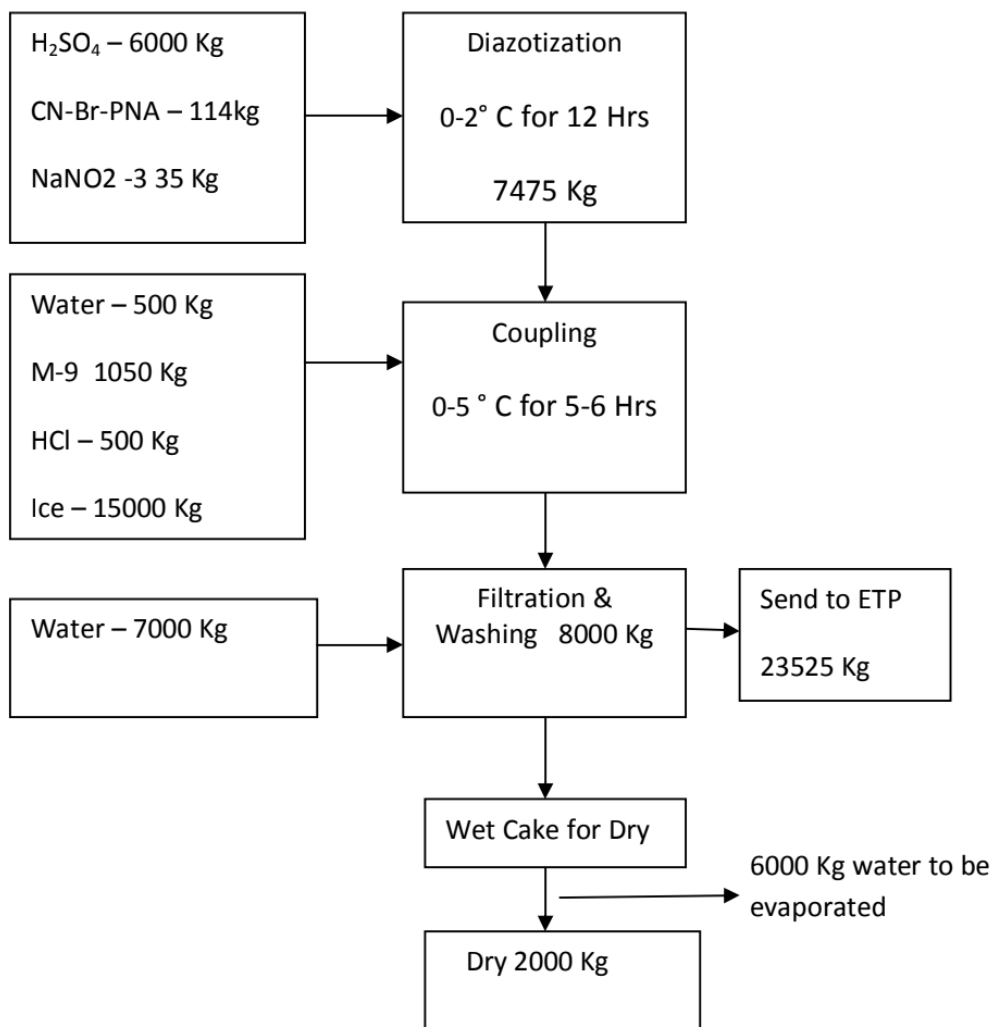
▼ 6. Annex

Detailed LCI for disperse dye has been given as example below

From a dyestuff manufacturer, the mass balance and production details has been obtained as shown below.

The production process follows a typical azo dye synthesis route involving diazotization, coupling, separation, and drying. Initially, the aromatic amine reacts with sodium nitrite in the presence of sulfuric acid at low temperatures (0–2 °C) to form a diazonium intermediate, requiring strict temperature control over ~12 hours. This intermediate is then coupled with the dye precursor (M-9) in an acidic medium under similarly low temperatures (0–5 °C), with significant use of ice for

cooling. The resulting slurry undergoes filtration and extensive washing, generating large volumes of wastewater sent to the effluent treatment plant. The product is recovered as a wet cake, which is subsequently dried by evaporating substantial moisture (~6000 kg water) to yield the final dry dye (~2000 kg), making the process both water- and energy-intensive.



Based on the above route, the following LCI has been developed.

Flow	Amount	Unit	Provider	Description
2-nitroaniline	0.525	kg	2-nitroaniline production 2-	The coupling intermediate was

			nitroaniline Cutoff, U - RER	modeled using nitro-aniline production as a proxy, as both represent aromatic amine intermediates used in azo dye synthesis with comparable nitration-based manufacturing chemistry. Differences in substitution introduce moderate uncertainty.
cooling energy	3.29	MJ	market for cooling energy cooling energy Cutoff, U - GLO	The cooling energy required to convert 7.5 kg of liquid water at an assumed initial temperature of 25 °C into ice at 0 °C was estimated by summing sensible and latent heat removal. First, the water is cooled to 0 °C using the specific heat capacity of water (4.18 kJ/kg·°C), giving about 272 kJ. Freezing then requires removal of latent heat of fusion (334 kJ/kg), adding roughly 868 kJ.

				The total thermodynamic cooling load is therefore , equivalent to 3.29 MJ. This represents the theoretical minimum heat removal
electricity, medium voltage	0.41596	kWh	market group for electricity, medium voltage electricity, medium voltage Cutoff, U - RER	Agitation electricity for diazotization, coupling, was estimated using a plaster mixing energy proxy (0.08 kWh/kg per stage) to represent mechanical stirring in aqueous batch reactors, introducing moderate uncertainty but preserving realistic energy magnitude. Also electricity consumption for spray drying was also included similar to thermal energy calculation. For filtration - ultrafiltration electricity

				consumption was used
heat, district or industrial, natural gas	15.72	MJ	market for heat, district or industrial, natural gas heat, district or industrial, natural gas Cutoff, U - RER without CH	Spray drying energy was modeled using an industrial milk spray-drying dataset expressed per kilogram of water evaporated. The dataset was scaled by the process-specific moisture removal (3 kg water per kg dye), assuming comparable thermodynamic drying behavior. Since spray drying energy is dominated by water evaporation, this proxy provides a reasonable approximation in the absence of plant-specific data.
hydrochloric acid, without water, in 30% solution state	0.25	kg	market for hydrochloric acid, without water, in 30% solution state hydrochloric acid, without water, in 30% solution state Cutoff, U - RER	

o-phenylene diamine	0.057	kg	o-phenylene diamine production o-phenylene diamine Cutoff, U - GLO	The MUA aromatic amine intermediate was modeled using o-phenylenediamine production as a proxy, as both represent aromatic amine fine chemicals produced via comparable nitration–reduction synthesis routes typical of dye intermediates. Some uncertainty remains due to differing substitution chemistry.
sodium nitrite	0.0175	kg	market for sodium nitrite sodium nitrite Cutoff, U - RER	
sulfuric acid	3	kg	market for sulfuric acid sulfuric acid Cutoff, U - RER	
tap water	11	kg	market for tap water tap water Cutoff, U - RER without CH	7.5 kg for ice conversion and 3.5 kg as water
Flow	Amount	Unit	Provider	Description
Disperse dye	1	kg		
wastewater, average	11.7625	l	market for wastewater,	

			average wastewater, average Cutoff, U - RER without CH	
Water	3	I		